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Duration	5 mins 29 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is the last element in the array

c. When the item is not the array at all

d. When the item is somewhere in the middle of the array ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. The list must be sorted

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. There should be direct access to the middle element in any sublist

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓

b. When the list is unsorted

c. When the list contains duplicate elements

d. When the list is very small

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The first element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search divides the list into halves.
c. Linear search is more efficient than binary search on large lists.
d. Linear search requires the list to be sorted.

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O👎
c. O(1)
d. O(log n) ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 94 and 99
c. 89 and 94
d. 90 and 99 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. log n
- d. n/2

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search stops immediately ✓
- c. The search continues until the end of the list
- d. The search starts over from the beginning

Question **10**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Binary search
- b. Jump search
- c. Linear search ✓
- d. Hashing

Question **11**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the left sublist
- b. The search continues in the right sublist
- c. The search ends successfully ✓
- d. The list is sorted

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Deleting elements from a list
c. Determining whether an element is present in a list ✓
d. Sorting elements in a list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Linear search ✓
c. Merge search
d. Bubble search

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. n/2
d. 1